

Lecture 1

4 main groups:

Technological Networks

Physical infrastructure (built over decades)
Backbone of modern societies: road and other transportation networks
power grids
telephone network
internet

Social Networks

Vertices/nodes = people, edges = social interaction

In sociology: vertices = actors, edges = ties

Social networks != twitter, facebook

Information Network

Data items linked to each other: relational databases
scientific publications linking through citations

WWW: nodes = webpages, edges = hyperlinks
one direction only
link $\rightarrow$ page
 $\therefore$ WWW is a directed network

Biological Network

Metabolic networks
Protein-protein interaction networks
genetic regulation networks

} molecular level patterns of interaction and control mechanisms in the biological cell

Ecological Networks

A neural network: vertices = neurons, edges: excitatory inputs
inhibitory inputs

Network Measures

Centrality - how important a particular edge or vertex is

$\rightarrow$ degree - the number of edges it is part of

Relevant measurements depend on the type of network

Network Concepts

Small-world effect: how far apart two randomly chosen points are
distance = number of edges needed along a path between
size degrees of separation

Course Contents:

1. Graphs and Graph Theory
2. Centrality Measures
3. Random Graphs
4. Small Worlds
5. Power Laws and Scale-Free Graphs
6. Models of Growing Graphs

A network perspective can simplify internal structure patterns and provide rich feedback effects on:

- social dynamics
- economic interaction
- online information
- designed technology

Lecture 2

a graph can serve as a mathematical model of a network.

- adjacencies?

Defn. A (simple) graph is a pair $G = (X, E)$, consisting of a (finite) set X of objects/nodes/vertices/points and $E \subseteq \binom{X}{2}$ of links/edges

$$\binom{X}{2} = \binom{n}{2} = \frac{1}{2} n(n-1) \text{ elements}$$

$$\text{Order} = n = |X|$$

$$\text{Size} = m = |E| \leq \binom{n}{2}$$

Each node has a list of neighbors, the nodes it is directly connected to

Degree number of neighbors of a node

Complete graph: $E = \binom{X}{2} = \binom{n}{2}$

Petersen graph: $X = 10$, $E = 15$.

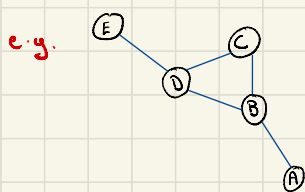
The complement of complete graph K_5
i.e. as the graph with vertex set $X = \{0, 1, 2, 3, 4\}$ (the edge set of K_5)

with an edge between $x, y \in X$ when $x \cap y = \emptyset$

Cycle graph

Lecture 3

Adjacency Matrix: square matrix, entry 1 or 0 in row i , column j , if the vertices are joined by an edge or not



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Symmetric: $a_{ij} = a_{ji}$ for all i, j

Degree

$$\text{degree of vertex } x_i = k_i = \sum_j a_{ij} = \sum_j a_{ji}$$

sum of row sum of column

$$\text{degree of all vertices} = \sum_i k_i = \sum_{i,j} a_{ij} = 2m$$

if edges = odd

The average degree of the nodes depends only on n and m .

$$= \frac{1}{n} \sum_i k_i = \frac{2m}{n}$$

Handshake lemma: any simple graph has an even number of nodes of odd degree

Graphs are Relations

Defn. A relation from a set X to a set Y is (nothing but) a subset R of the Cartesian product $X \times Y = \{(x, y) : x \in X, y \in Y\}$.

We say that $x \in X$ is R -related to $y \in Y$ whenever $(x, y) \in R$ and then write xRy

Relation adjacency matrix: $R \subseteq X \times Y$

row for each $x \in X$
column for each $y \in Y$
1 if xRy , 0 otherwise

Homogeneous relation: $Y = X$.

properties:
R: reflexive if xRx for all $x \in X$ $\checkmark 2 \leq 2, 2 \leq 2 \Rightarrow$ R.S.T. = equivalence relation

S: symmetric if xRy implies yRx for all $x, y \in X$ $\times 2 \leq 3 \neq 3 \leq 2$ partitions X into (mutually disjoint) parts

T: transitive if xRy and yRz imply that xRz for all $x, y, z \in X$ $\checkmark 2 \leq 3, 3 \leq 5 \Rightarrow 2 \leq 5$ equivalence classes

I: irreflexive if not xRx for all $x \in X$ $\checkmark \Rightarrow$ R.A.T. = partial order

A: antisymmetric if xRy and yRx imply that $x = y$ for all $x, y \in X$ $\checkmark$ e.g. divides partial order of natural numbers subsets of a set

$\Rightarrow$ Simple graph is a symmetric, irreflexive relation on X . $\Rightarrow$ Directed graph is irreflexive if no loops
A antisymmetric if every edge has a unique direction

Composition and Adjacency Matrices

Composite

Relation: xRy , ySz then $x(R \circ S)z$ if there is an element $y \in Y$ such that xRy, yRz
eg. grandfather relationships are composites of father relationships

Composition

Adjacency Matrix: is essentially the matrix product of the adjacency matrices of the individual relations R and S .

If $A = (a_{ij})$ is $\text{adj}(R)$ and $B = (b_{jk})$ is $\text{adj}(S)$

then the i, k -entry of the product AB is $(AB)_{ik} = \sum_j a_{ij} b_{jk}$

which is exactly the number of elements $y \in Y$ such that $x_i R y$ and $y S z_k$

All it needs for x_i to be $(R \circ S)$ -related to z_k is this number to be at least 1.

$\therefore$ replacing all non-zero entries in the product matrix AB with 1 yields the adjacency matrix of $R \circ S$

Lecture 4: Bipartite Graphs and Projections

Bipartite: a (simple) graph $G=(X,E)$ is called bipartite if the vertex set X is a disjoint union of two sets B (of black nodes) and W (of white nodes) so that each edge in E links a black vertex with a white vertex.

Vertex-Coloring coloring nodes so that any two connected nodes have different colours

G is bipartite = G is 2-colorable = each cycle in G has even length

Affiliation Network: people = black nodes
attributes = white nodes e.g. common interests, workplaces...
Edges connect people with their attributes

Bipartite Adjacency:

$$A = \begin{pmatrix} 0 & C \\ C^T & 0 \end{pmatrix}$$

no edges between vertices of the same color

Projection Adjacency:

$$A^T A = A A^T = \begin{pmatrix} C C^T & 0 \\ 0 & C^T C \end{pmatrix}$$

adjacency matrix for projection onto first set of nodes

Lecture 5

paths in a $g = (X, E)$ is a sequence of nodes where consecutive nodes are linked by an edge in E

simple path if no nodes are repeated

length number of edges = nodes - 1

cycle path of length at least 3 with first and last node eg. (H, J, K, L, H)
in a simple graph: for any two nodes, at least two different ways between

connected ^(simple) graph with path between every pair of nodes

connected components not connected graph broken down

path is reflexive
symmetric
transitive

acyclic graph contains no cycles

trees simple graph that's connected, acyclic

g is a tree = g is connected and $m = n - 1$ = g is minimally connected = g is acyclic and $m = n - 1$ = g is maximally acyclic
i.e. acyclic and connected

removing any edge will disconnect g

adding any edge will introduce a cycle

Cayley's Formula There are exactly n^{n-2} distinct (labelled) trees on the n -element vertex set.

$X = \{0, 1, 2, \dots, n-1\}$ if $n > 1$

Lecture 8 Degree and Eigenvector Centrality

degree centrality : $g = (X, E)$ with $X = \{0, \dots, n-1\}$ and $A = (a_{ij})$
$$c_i^D = k_i = \sum_j a_{ij}$$

normalised degree centrality :
$$c_i^D = \frac{k_i}{n-1} = \frac{c_i^D}{n-1}$$
 degree centrality divided by potential number of neighbours

eigenvector centrality
$$\lambda c_i^E = \sum_j a_{ij} c_j^E \Rightarrow \lambda c^E = A c^E$$

c^E ? λ ?

find characteristic polynomial $p_A(x)$ of A (as determinant $xI - A$)

calculate roots λ of $p_A(x)$ such that $p_A(\lambda) = 0$

find nontrivial solutions $(\lambda I - A)c = 0$

Perron - Frobenius Theorem:

Square, nonnegative, irreducible matrix

A has a real eigenvalue $\lambda > 0$ with $\lambda \geq |\lambda'|$ for all eigenvalues λ' of A
 λ is a simple root

there is a λ -eigenvector v with $v > 0$

$$\text{e.g. } \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

all a_{ij} $A = a(i,j)$ are positive

eigenvalue of 4 $> |\lambda'|$ of all other eigenvalues (here $\lambda' = 1$)

eigenvector corresponding to $\lambda = 4$ has all positive elements.

closeness centrality

betweenness centrality

